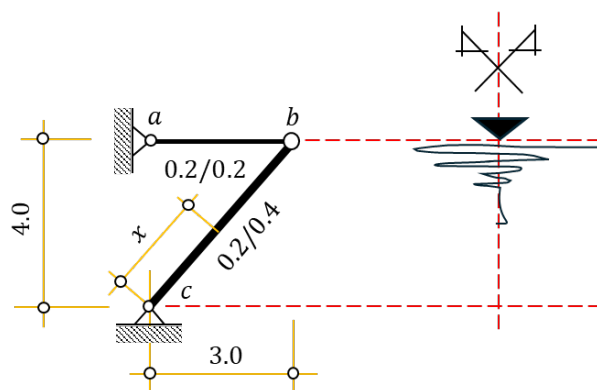


1: За систем на скици услед дејства течности $\gamma = 10 \text{ kN/m}^3$ срачунати и нацртати:

- статичку и кинематичку класификацију,
- реакције у ослоњцима и силе у пресецима,
- на делу "с-б", положај попречног пресека са максималном вредности М савијања.

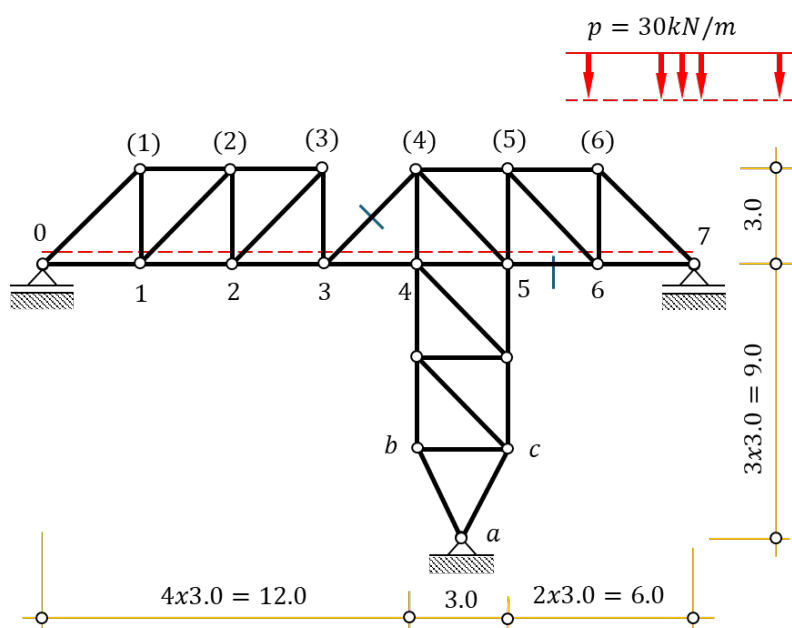
Подаци: $E = 3 \cdot 10^7 \frac{\text{kN}}{\text{m}^2}$,



2: За решеткасти носач приказан на скици срачунати и нацртати:

- утицајне линије за силе у штаповима "3 - (4)", и "5 - 6".
- меродавне или опасне положаје за покретно једнакоподељено оптерећење, односно максималне и минималне вредности сила у штаповима "3 - (4)", и "5 - 6".
- за меродаван положај оптерећења који изазива максималну силу у штапу "3 - (4)", коришћењем аналитичких израза срачунати силе у штаповима решеткастог носача.

Подаци: $E = 3 \cdot 10^7 \frac{\text{kN}}{\text{m}^2}$, штапови су: $b/h = 20/20 \text{ cm}$.



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$G = G$ $\Rightarrow$ СТАТИЧКИ ОДРЕЂЕН
КИНЕМАТИЧКИ
ПРОСТО СТАБИЛАН

10 ВАРУЖАТА

№ 114444

$$5 + 2 = 2.5 - 3$$

$$7 = 7$$

OK

20 НАЧМН

$$2.3 = 3.3 - 3$$

$$6 = 6$$

OK

2^o BAPUJATA

10 HXCH

$$\downarrow$$
$$6 + 3 = 2 \cdot 6 - 3$$

$$g = g$$

OK

2P #XMH

$$2.3 = 3.3 - 3$$

$$6 = 6$$

OK

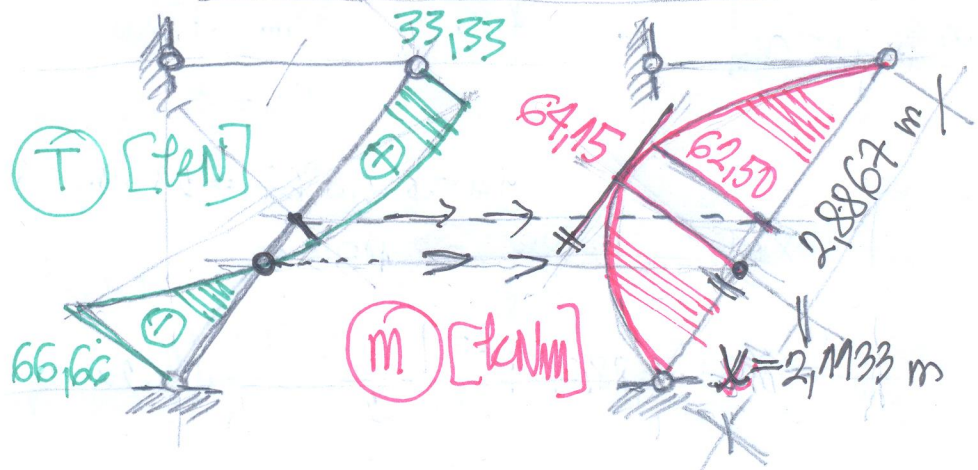
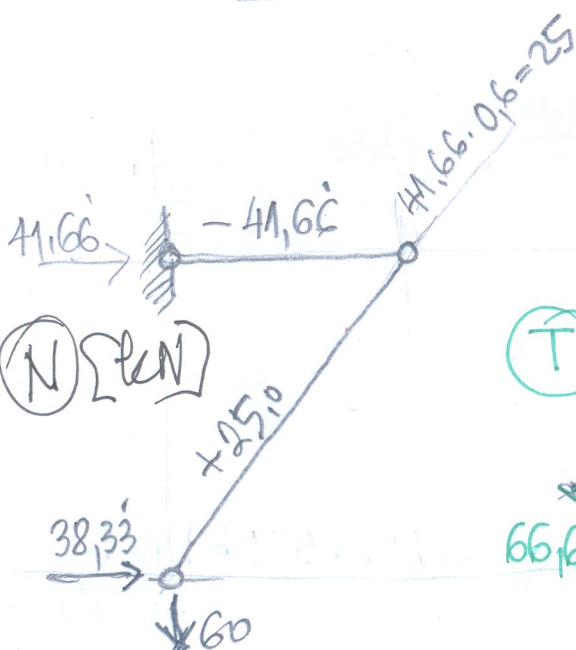
* УНУТРАШЊЕ КИНЕМАТИЧКИ ПРОСТО СТАБИЛАН ПОСЛУЖИТАТО.

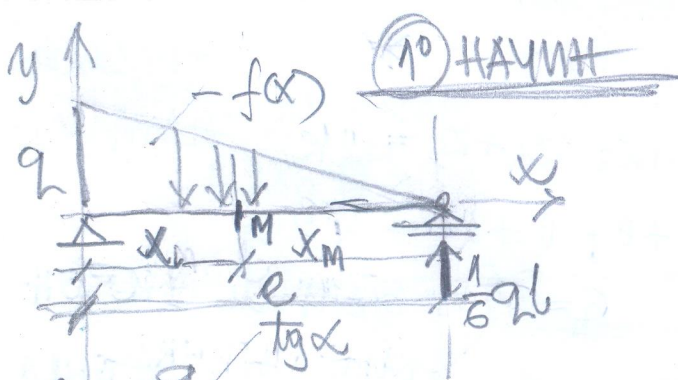
$$\Sigma H = 0: H_a + H_c - 100 \cdot 0,8 = 0$$

$$\Sigma V = 0: V_c + \cancel{V_a} + 100 \cdot 0,6 = 0$$

$$\Sigma \hat{M}_C = 0: -H_A \cdot 4,0 + 100 \cdot 0,8 \cdot 0,8 \cdot 1,66 + 100 \cdot 0,6 \cdot 0,6 \cdot 1,66 = 0$$

Результат: $H_a = 41,66 \text{ кН}$

$$H_c = 38,33 \text{ kN}$$
$$Y_c = -60 \text{ kN}$$




$$f(x) = \frac{q}{l} \cdot x$$

$$T(x) = 0 - \text{yaro 3a } \max M(x)$$

$$T(x) = \frac{1}{6} q \cdot l - \int_0^l \frac{q}{l} \cdot x_m dx$$

$$T(x) = \frac{q \cdot l}{6} - \frac{q}{l} \frac{x_m^2}{2} + c \quad \text{3a } x=0$$

$$T(x) = -\frac{1}{2l} q \cdot x_m^2 + \frac{1}{6} q \cdot l \quad \text{3a } T(x) = 0$$

$$x_m^2 = \frac{\frac{1}{6} q \cdot l}{\frac{1}{2l} q} = \frac{1}{3} l^2$$

$$\Rightarrow x_m = \sqrt{\frac{1}{3} l^2} = \frac{1}{\sqrt{3}} l = \frac{1}{\sqrt{3}} \cdot 5,0 = 2,8867 \text{ m}$$

$$x = l - x_m = 5 - 2,8867 = 2,1133 \text{ m}$$

(2) HAYM

$$N_a^{\text{rope}} = 41,66 \cdot 0,6 = 25 \text{ kN}$$

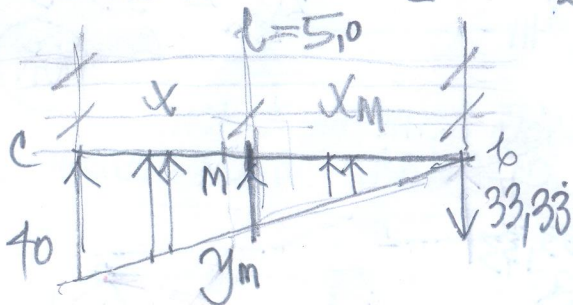
$$N_c^{\text{rope}} = -38,33 \cdot 0,6 + 60 \cdot 0,8 = 25 \text{ kN}$$

$$T_a^{\text{rope}} = 41,66 \cdot 0,8 = 33,33 \text{ kN}$$

$$T_c^{\text{rope}} = -38,33 \cdot 0,8 - 60 \cdot 0,6 = -66,66 \text{ kN}$$

$$T(\frac{1}{2}l) = 41,66 \cdot 0,8 - \frac{20 \cdot 2,5}{2} = 8,33 \text{ kN} \quad 0,833$$

$$M(\frac{1}{2}l) = 33,33 \cdot \frac{5}{2} + \frac{20 \cdot 2,5}{2} \cdot \left(\frac{2}{3} \cdot 2,5\right) = 62,5 \text{ kNm} \quad \text{TO TMSJE } \max M$$



$$x_m : y_m = 5 : 40$$

$$5y_m = 40x_m$$

$$y_m = 8x_m$$

$$y_m = 8 \cdot 2,8867 = 23,094 \text{ kN}$$

$$x = l - x_m = 5,0 - 2,8867 = 2,1133 \text{ m}$$

$$(2) \text{ yaro: } T_m = 0$$

$$33,33 - \frac{x_m \cdot y_m}{2} = 0$$

$$\frac{1}{2} x_m \cdot y_m = 33,33 \quad / \cdot 2$$

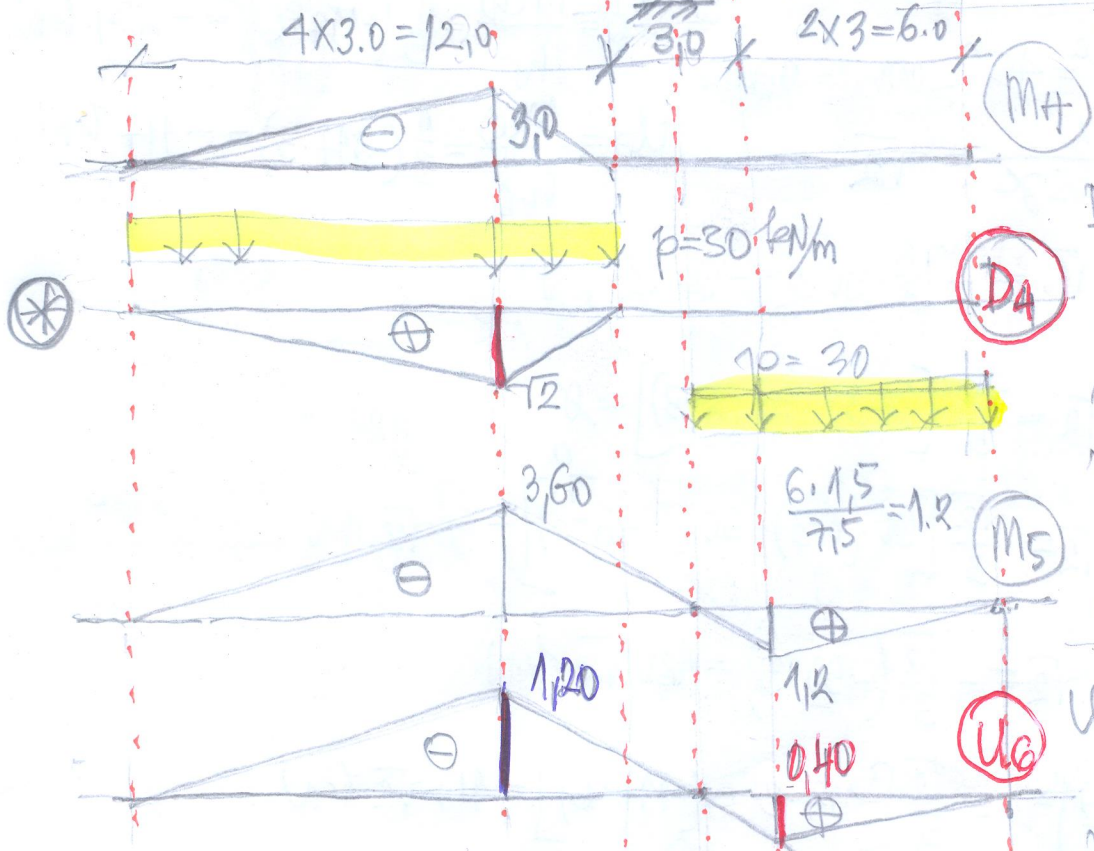
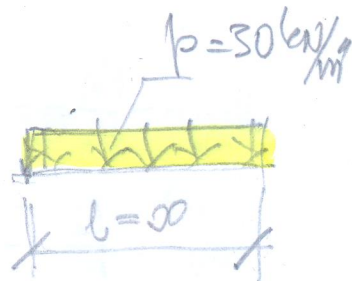
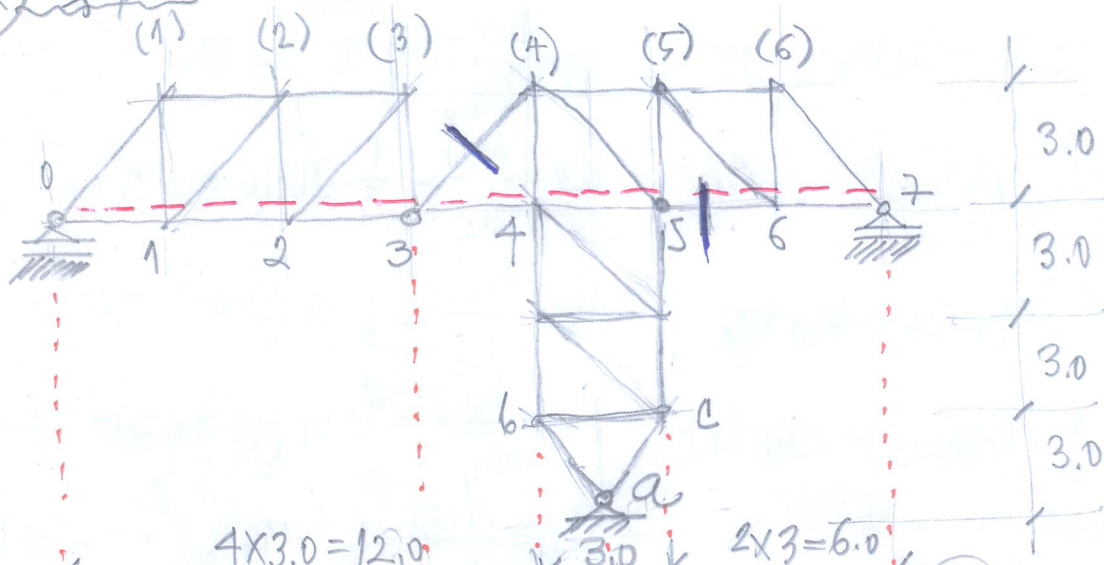
$$8x_m^2 = 66,66$$

$$x_m^2 = \frac{66,66}{8} \Rightarrow x_m = 2,8867$$

$$M_m = 33,33 \cdot 2,8867 + \frac{23,094 \cdot 2,8867}{2} \cdot \left(\frac{2}{3} \cdot 2,8867\right) = 64,149 \approx 64,15 \text{ kNm}$$

OK! 😊

② 3AMATAK (1)



$$D_4 = \left(\frac{M_3}{h_3} - \frac{M_4}{h_4} \right) \cdot \frac{1}{\cos \theta_4}$$

$$D_4 = -\frac{1}{3}M_4\sqrt{2} = -\frac{\sqrt{2}}{3}M_4$$

$$= -\frac{\sqrt{2}}{3} \cdot 3 = -\sqrt{2}$$

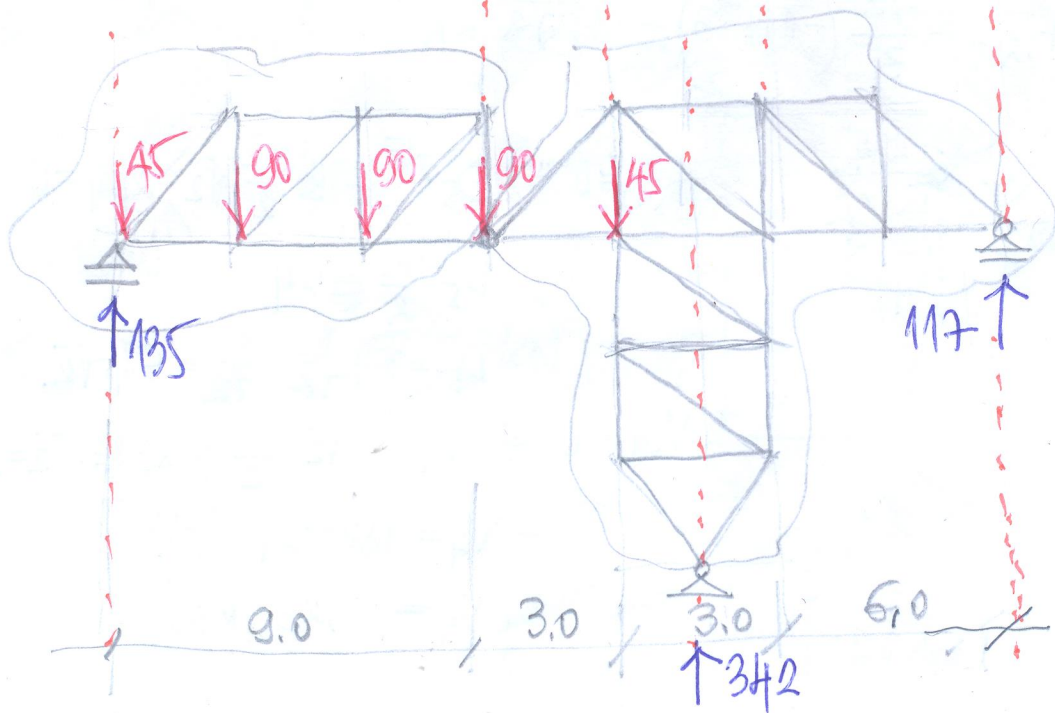
$$\min D_4 = 0$$

$$\begin{aligned} \max D_4 &= p \cdot F = \\ &= 30 \cdot \frac{\sqrt{2}}{2} \cdot 12 = \\ \max D_4 &= 254,558 \text{ kN} \end{aligned}$$

$$U_6 = \frac{M(s)}{s} = \frac{1}{s} \cdot M(s)$$

$$\begin{aligned} \min U_6 &= p \cdot \bar{F} = \\ &= 30 \cdot \left(\frac{1,2}{2} \cdot 13,5 \right) = \\ \min U_6 &= 243,0 \text{ kN.} \end{aligned}$$

$$\begin{aligned} \max U_0 &= p \cdot F^+ = \\ &= 30 \cdot \frac{1}{2} \cdot 0,475 = \\ \max U_0 &= 45 \text{ kN} \end{aligned}$$



17.04.2026.

2. Wavelength

$$D_{(2)} = -\frac{M_1}{h_1} = -\frac{1}{3} 90 \cdot 3 = -90 \text{ kN}$$

$$D_{(3)} = -\frac{M_2}{h_2} = -\frac{1}{3} (90 \cdot 6 - 90 \cdot 3) = -90 \text{ kN}$$

$$D_{(6)} = -\frac{M_6}{h_6} = -\frac{1}{3} (-117 \cdot 3) = 117 \text{ kN}$$

$$D_{(5)} = -\frac{M_5}{h_5} = -\frac{1}{3} \cdot (-117 \cdot 6) = 234 \text{ kN}$$

$$U_1 = \frac{M_{(1)}}{h_1} = \frac{1}{3} \cdot 90 \cdot 3 = 90 \text{ kN}$$

$$U_2 = \frac{M_{(2)}}{h_2} = \frac{1}{3} \cdot 90 \cdot (6-3) = 90 \text{ kN}$$

$$U_3 = \frac{M_{(3)}}{h_3} = \frac{1}{3} 90 \cdot (9-6-3) = 0$$

$$U_4 = \frac{M_{(4)}}{h_4} = \frac{1}{3} 90 (12-9-6-3) = -180 \text{ kN}$$

$$U_6 = \frac{M_{(5)}}{h_5} = \frac{1}{3} (-117 \cdot 6) = -234 \text{ kN}$$

$$U_7 = \frac{M_{(6)}}{h_6} = \frac{1}{3} (-117 \cdot 3) = -117 \text{ kN}$$

$$\theta^k = 45^\circ \quad \cos \theta^k = 0.707$$

$$\frac{1}{\cos \theta^k} = \sqrt{2}$$

$$D_{(1)} = \left(\frac{M_0}{h_0} - \frac{M_{(1)}}{h_1} \right) \sqrt{2} = -\frac{\sqrt{2}}{3} \cdot 90 \cdot 3 = -90\sqrt{2} \text{ kN}$$

$$D_{(2)} = \left(\frac{M_1}{h_1} - \frac{M_{(2)}}{h_2} \right) \sqrt{2} = \frac{\sqrt{2}}{3} [90 \cdot 3 - 90 \cdot (6-3)] = 0$$

$$D_{(3)} = \left(\frac{M_2}{h_2} - \frac{M_{(3)}}{h_3} \right) \sqrt{2} = \frac{\sqrt{2}}{3} [90 \cdot (6-3) - 90 \cdot (9-6-3)] = 90\sqrt{2} \text{ kN}$$

$$D_{(4)} = \left(\frac{M_3}{h_3} - \frac{M_{(4)}}{h_4} \right) \sqrt{2} = -\frac{\sqrt{2}}{3} (-90 \cdot 3) = 180\sqrt{2} \text{ kN}$$

$$D_6 = \left(\frac{M_6}{h_6} - \frac{M_{(5)}}{h_5} \right) \sqrt{2} = \frac{\sqrt{2}}{3} [-117 \cdot 3 - (-117 \cdot 6)] = 117\sqrt{2} \text{ kN}$$

$$D_7 = \left(\frac{M_7}{h_7} - \frac{M_{(6)}}{h_6} \right) \sqrt{2} = -\frac{\sqrt{2}}{3} (-117 \cdot 3) = 117\sqrt{2} \text{ kN}$$

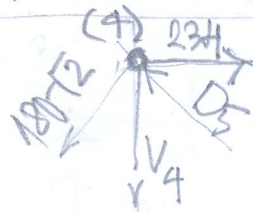
$$V_1 = 90 \text{ kN}$$

$$V_2 = 0$$

$$V_3 = -90 \text{ kN}$$

$$V_6 = -117 \text{ kN}$$

$$V_5 = -117 \text{ kN}$$



$$\sum H_{(4)} = 0 \Rightarrow D_5 \cdot \frac{\sqrt{2}}{2} + 180 \frac{\sqrt{2}}{2} \sqrt{2} - 234 = 0$$

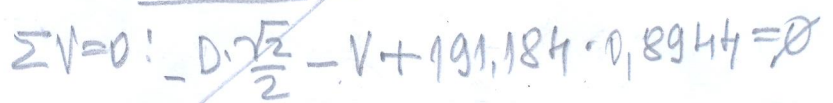
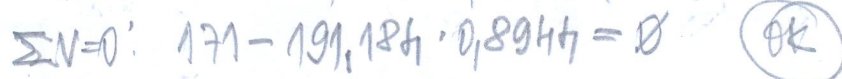
$$D_5 \frac{\sqrt{2}}{2} = 54$$

$$D_5 = 54 \cdot \frac{2}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = 54\sqrt{2}$$

$$\sum V_{(4)} = 0 \Rightarrow V_4 - 54\sqrt{2} \cdot \frac{\sqrt{2}}{2} + 180 \frac{\sqrt{2}}{2} \cdot \sqrt{2} = 0$$

$$-V_4 = 180 - 54 (-1)$$

$$V_4 = -126 \text{ kN}$$



$$V = 171 \text{ kN (upward calc)}$$

